

Ahmed Fakhreldin

Integration Engineer

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PROFESSIONAL SUMMARY

Associate Integration Engineer with hands-on experience in enterprise middleware solutions and banking systems integration. Specialized in modernizing legacy IBM App Connect Enterprise services, delivering robust API integrations, and optimizing middleware architectures for high-volume financial transactions. Proven track record in improving system reliability, maintainability, and performance in mission-critical banking environments.

PROFESSIONAL EXPERIENCE

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Associate Integration Engineer

Cairo, Egypt

Jun 2025 – Present

- **Bank Al-Bilad Integration Modernization:** Led the modernization of complex legacy IBM App Connect Enterprise (ACE) services within IBM Cloud Pak for Integration (CP4I) platform, transforming monolithic integration flows into modular, maintainable service-oriented integration components, resulting in measurable improvements in transaction performance.
- **API Lifecycle Automation:** Engineered end-to-end automation framework for IBM API Connect using Bash scripting and APIC CLI, managing large-scale API deployments with template-based configuration and Git-driven incremental builds, reducing deployment cycle time from Days to minutes (99% time reduction) while eliminating manual configuration errors across development and production environments.
- Deployed and managed integration solutions on OpenShift Container Platform, leveraging CP4I's cloud-native architecture for containerized deployment, scaling, and orchestration of integration workloads in enterprise Kubernetes environments.
- Designed and implemented RESTful APIs and message-driven integrations using IBM ACE/IIB, supporting critical banking operations including payment processing, account management, and inter-system communications.
- Enhanced middleware performance by refactoring integration flows, implementing error handling strategies, and establishing monitoring frameworks, reducing system downtime by 25% and improving overall reliability.
- Collaborated with cross-functional teams including backend developers, database administrators, and business analysts to gather requirements, design integration solutions, and ensure seamless data flow across banking systems.
- Developed and maintained technical documentation for integration patterns, API specifications, and deployment procedures, facilitating knowledge transfer and ensuring consistent implementation standards.

TECHNICAL SKILLS

Integration Platforms: IBM Cloud Pak for Integration (CP4I), IBM App Connect Enterprise (ACE), IBM Integration Bus (IIB), IBM MQ, Event Streams, IBM API Connect, Datapower, REST APIs, SOAP Web Services.

Programming Languages: ESQL, Java, C#, JavaScript, Python, TypeScript, C++, XML.

Backend Technologies: ASP.NET Core, Entity Framework Core, Node.js, Express.js.

Frontend Frameworks: Angular, React.js, HTML5, CSS3.

Databases: Oracle DB, SQL Server, MongoDB, Neo4j, Database Design & Optimization.

Cloud & Container Platforms: OpenShift Container Platform, Kubernetes, Docker, CP4I Deployment.

Tools & Methodologies: Git, Visual Studio Code, Agile/Scrum, CI/CD, Apache Kafka, API Testing (Postman / SoapUI).

EDUCATION

University of Ain Shams

Bachelor of Computer and Systems Engineering (GPA: 3.22/4.00, Very Good)

Cairo, Egypt

Sept 2018 – Jun 2023

- **Relevant Coursework:** Distributed Systems, Data Structures & Algorithms, Software Engineering, Database Systems, Operating Systems, Design Patterns, Parallel & Distributed Systems, Artificial Intelligence

TECHNICAL PROJECTS

Smart Examination System | *Angular, Node.js, MongoDB, Express, TypeScript, ML*

- Architected and built a full-stack examination platform with RESTful API backend, serving 500+ concurrent users with responsive Angular frontend and Node.js/Express backend infrastructure.
- Implemented streamlined MongoDB data schemas and caching strategies, achieving 60% faster query response times for high-volume assessment data retrieval and storage operations.
- Integrated machine learning models (T5 Transformer, Word2Vec, Random Forest) for intelligent question generation, and third-party APIs for text-to-speech and speech recognition, enabling multi-modal assessment capabilities.

- Authored IEEE-standard research paper on test generation methodologies, synthesizing findings from 5+ seminal studies in educational technology and natural language processing.
- XML Processing Desktop Application** | *C++, Data Structures, Algorithms*
- Engineered a high-performance XML editor implementing 5 core functionalities: XML-to-JSON conversion, data compression, syntax validation, error correction, and file prettification.
 - Crafted and refined custom algorithms for efficient tree-based data structure manipulation, achieving $O(n \log n)$ complexity for parsing and transformation operations on large XML files.
- Othello AI Game Player** | *Python, Minimax Algorithm, Alpha-Beta Pruning, PyGame*
- Created an intelligent game AI using Minimax algorithm with alpha-beta pruning optimization, analyzing 500+ potential moves per turn to determine optimal strategic gameplay decisions.
 - Formulated sophisticated heuristic evaluation functions considering board position, mobility, and control patterns, achieving 85% win rate against intermediate human players.
 - Built interactive PyGame-based user interface enabling seamless human-AI gameplay with real-time move visualization and game state management.

MILITARY SERVICE

Egyptian Armed Forces	Cairo, Egypt
<i>Completed Mandatory Military Service with Honors</i>	<i>Oct 2023 – Dec 2024</i>
• Demonstrated leadership and operational excellence in high-pressure environments, coordinating activities for 60+ personnel while executing complex operational tasks with precision and strategic decision-making.	

CERTIFICATIONS

• Foundation of Generative AI	<i>Jan 2025</i>	• MEAN Stack Development (ITI)	<i>Sep 2022</i>
• IBM Event Streams Developer Essentials	<i>Nov 2024</i>	• Algorithmic Toolbox (UC San Diego)	<i>Sep 2021</i>
• IBM MQ Developer Essentials	<i>Oct 2024</i>	• AI For Everyone (Coursera)	<i>Jul 2021</i>
• Professional Frontend Development	<i>Dec 2022</i>	• Python for Everybody (Coursera)	<i>Sep 2020</i>

LANGUAGE PROFICIENCY

• Arabic: Native (C2)	• English: Proficient (C2)	• German: Beginner (A1)
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